Tutorial Sheet 4

IIT Delhi – Abu Dhabi

Course Code: AENL226

Course Name: Power Electronics and Power Systems

1. The voltage across and current through a load are shown in Fig. 1(a) and (b). Both the waveforms are periodic with T=20 ms.

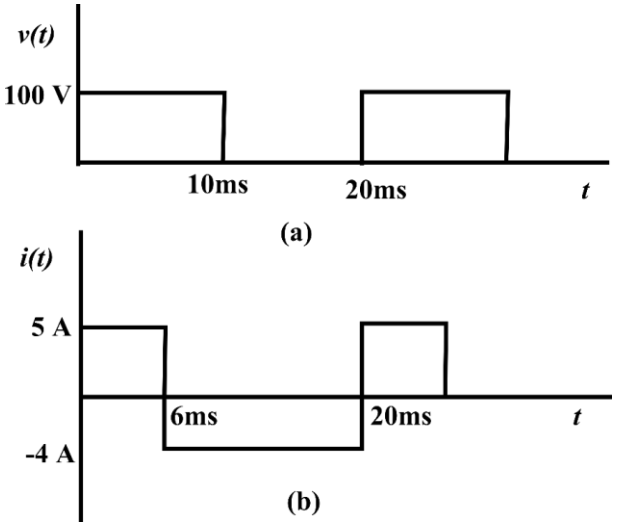
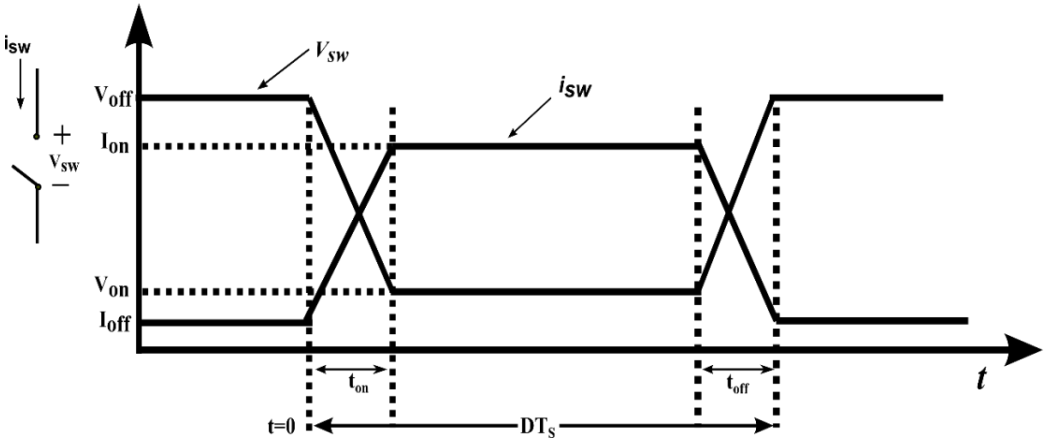


Fig.1

Determine (a) the instantaneous power, (b) the average power, and (c) the energy absorbed by the load in each period.

Ans (a) 500W,-400W, 0W (b) 70W (c) 1.4J

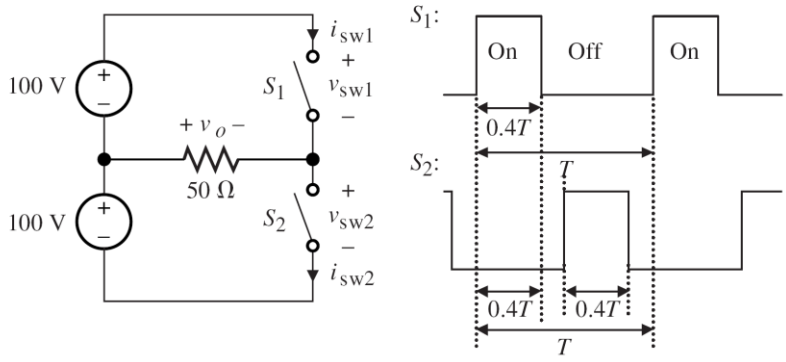
2. Consider a power switching device whose voltage and current waveforms are shown in Fig. 2. Assume V_{off} =100V, V_{on} = 0.7V, I_{on}=5 A, I_{off} =0.2 A, D =1, T_s =150µs, t_{on}= 5µs and t_{off} =10 µs.



(a) Find an expression for the instantaneous switching and conduction power losses in the device.
(b) Determine the energy dissipated in the device during one switching period.
(b)Determine the average power dissipated in the device during one switching period.

Ans: (b) 1722uJ (c) 11.5 W

3. Consider the two switch circuit shown in Figure, where S1 and S2 are assumed ideal.
(a) Sketch isw1, vsw1, isw2, vsw2.
(b) Find out the average output voltage.
(c) Find out the average power delivered to the load.



Ans: (b) 0 V (c) 160W

4. The power dissipated in a device is 20 W. The thermal resistance of the device is 1^0C/W. Assume the thermal interface material has thermal resistance of 1.5^0C/W. To maintain the junction temperature below 100^0C for 25^0C ambient temp, what should be the thermal resistance of the heat sink?

Ans: 1.25 ^{0}C /W

5. A thyristor has a full load average current rating of 65 A when the temperature of its heat sink is maintained at 65°C and the maximum permissible junction temperature is 120°C. The thermal resistance between the junction and the heat sink of the device is 0.92°C/W. Assume that the total internal power dissipation is forward conduction dissipation only. Determine:
a) Internal power dissipation in the thyristor.
b) The percentage increase in device current rating if the heat sink temperature is brought down to 55°C by efficient cooling.

Ans : (a) 59.78 W(b) 8.7 %